



Digital Float Control System

Version 1.0, April 15, 2025: First Official Release

Table of Contents

1. Introduction	2
2. System Overview	3
3. Installation and Setup	4
3.1. Physical Installation	4
3.1.1. Mounting the Control Unit	4
3.1.2. Connecting Motor Channels	4
3.1.3. Switch Inputs	4
3.1.4. Power Connection	4
3.2. Initial Configuration	4
4. Basic Operation	6
4.1. Normal Operation Mode	6
4.2. Operational States	6
4.3. Switching Procedure	6
5. LED Indicators	7
5.1. Landing Gear Indicators	7
5.2. Rudder Indicators	7
5.3. Custom Colors	7
6. Test Switch Functions	8
6.1. Momentary Press	8
6.2. Extended Press	8
7. Configuration Options	9
7.1. User Settings	9
7.2. Channel Settings	9
8. Command Protocol	10
8.1. Supported Commands	10
8.2. Command Format	10
9. Troubleshooting	11
9.1. Error Indicators	11
9.2. Common Issues	11
9.3. Firmware Update	11
10. Maintenance	12
10.1. Routine Inspection	12
10.2. Calibration	12
11. Technical Specifications	13
11.1. Electrical	13
11.2. Mechanical	13
11.3. Compliance Information	13
12. Warranty and Support	14

Appendix A: Command Reference	15
Appendix B: Glossary.....	16
Colophon.....	17

Digital Float Control System

User Manual

For Experimental and Special Category Aircraft

Version 1.0

Skymatik Aero

April 15, 2025

CONFIDENTIAL: This document contains proprietary information and is provided under a license agreement. It may not be copied or shared without written permission from Skymatik Aero.

Copyright © 2025 Skymatik Aero. All Rights Reserved.



The Digital Float Control System is designed for installation in Experimental, Amateur-Built, and Special Category aircraft only. This product is not FAA/EASA certified for installation in type-certificated aircraft.

Installation of this equipment in an aircraft is a major alteration and may require appropriate documentation and testing in accordance with applicable aviation regulations. Builders and owners are responsible for ensuring compliance with all applicable regulations regarding installation and operation of this equipment.

Chapter 1. Introduction

Thank you for choosing the Digital Float Control System. This manual provides detailed information on the installation, operation, and maintenance of your system. The Digital Float Control System is designed to provide reliable control of experimental aircraft landing gear and rudder motors, with visual status indication via LEDs.



This system is specifically designed for experimental and special category aircraft. It is not approved for installation in standard type-certificated aircraft under FAA Part 23, Part 25, Part 27, Part 29 or equivalent EASA regulations without appropriate field approval or STC (Supplemental Type Certificate).

Table 1. Document Information

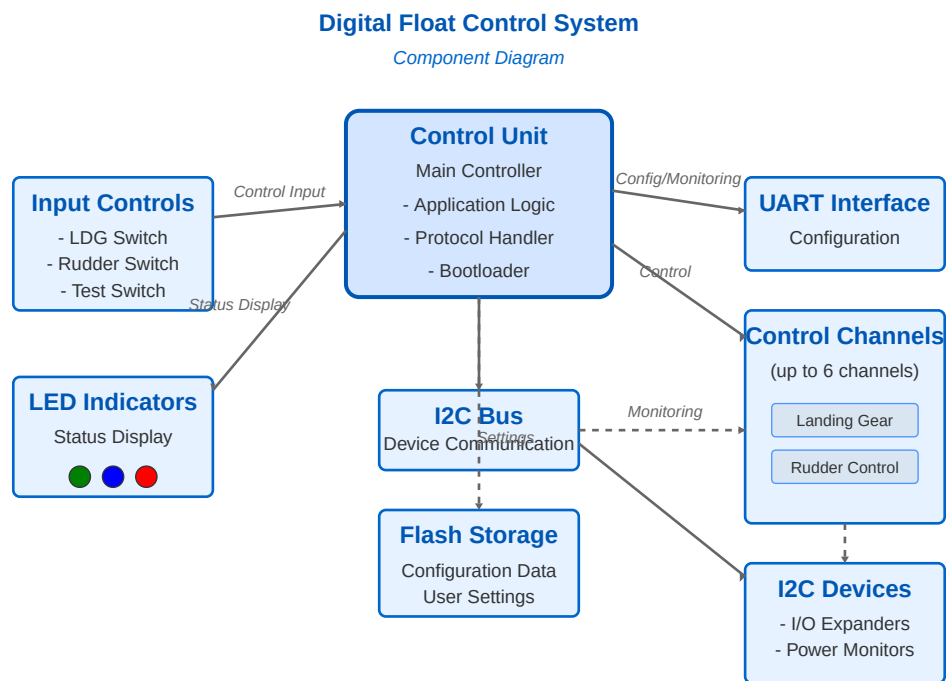
Item	Description
Document Version	1.0
Release Date	April 15, 2025
Document Status	First Official Release
Product	Digital Float Control System for Experimental Aircraft
Manufacturer	Skymatik Aero
Aircraft Applicability	Experimental, Amateur-Built, and Special Category Only

Chapter 2. System Overview

The Digital Float Control System consists of the following major components:

- **Control Unit:** Central processing module with firmware that manages all system functions
- **LED Indicators:** Multi-color status lights that display system state
- **Control Channels:** Up to 6 independent channels for motor control
- **Input Switches:** Landing gear and rudder position switches
- **Test Switch:** For diagnostics and configuration functions
- **Communication Port:** UART interface for configuration and diagnostics

Each control channel can be configured for either landing gear or rudder operation, with different behavior profiles for each type. This flexibility makes the system ideal for various experimental aircraft configurations.



© Skymatik Aero

Figure 1. System Block Diagram

Figure 1: Digital Float Control System Component Diagram

Chapter 3. Installation and Setup

3.1. Physical Installation



Installation in experimental aircraft should follow best practices for aviation electrical systems. While this product is designed for reliability, the builder/operator is responsible for ensuring proper installation, testing, and operation.

3.1.1. Mounting the Control Unit

- Mount the unit in a dry, accessible location using the four mounting holes
- Ensure adequate ventilation and protection from vibration
- Operating temperature range: -20°C to +70°C



Ensure proper cooling and ventilation for the control unit. Inadequate cooling may result in overheating and reduced reliability.

3.1.2. Connecting Motor Channels

- Connect control channels to the appropriate motors
- Observe proper polarity as indicated on the terminal labels
- Maximum load: 2A per channel at 28V DC



Exceeding the maximum load rating may result in damage to the control circuits. Always use appropriate fusing and protection circuits.

3.1.3. Switch Inputs

- Connect landing gear position switch to the LDG terminal
- Connect rudder position switch to the RUD terminal
- Switches should be dry contact type (no voltage)

3.1.4. Power Connection

- Connect power supply (24V DC nominal)
- Observe proper polarity
- System requires 2A minimum supply capacity

3.2. Initial Configuration

After installation, the system requires initial configuration:

1. Apply power to the unit
2. Connect to the communication port using a terminal program
 - Baud rate: 115200
 - Data bits: 8
 - Parity: None
 - Stop bits: 1
 - Flow control: None
3. Use the command protocol (see [Command Protocol](#)) to configure channel settings



Before beginning configuration, verify all electrical connections and ensure that the system is receiving proper power.

Chapter 4. Basic Operation

4.1. Normal Operation Mode

During normal operation, the system:

- Monitors the landing gear and rudder position switches
- Controls the appropriate motors based on switch positions
- Displays status via LED indicators
- Monitors voltage and current for fault detection

4.2. Operational States

Each channel can be in one of the following states:

UP

Channel is in the up position

DOWN

Channel is in the down position

MOVING

Channel is transitioning between positions

ERROR

Channel has detected a fault condition

4.3. Switching Procedure

To operate the landing gear:

1. Move the landing gear switch to the desired position
2. System will automatically control motors to achieve the position
3. LED indicators will show the current state
4. Motors will stop automatically when position is reached

To operate the rudder:

1. Move the rudder switch to the desired position
2. System will automatically control the rudder motor
3. LED indicators will show the current state
4. Motor will stop automatically when position is reached

Chapter 5. LED Indicators

The system uses multi-color LED indicators to show the status of each channel:

5.1. Landing Gear Indicators

- **Green** (default): Landing gear is down and locked
- **Blue** (default): Landing gear is up and locked
- **Blinking**: Landing gear is in motion
- **Blinking Red**: Error condition detected

5.2. Rudder Indicators

- **Yellow** (default): Rudder is in central position
- **Purple** (default): Rudder is in deflected position
- **Blinking**: Rudder is in motion
- **Blinking Red**: Error condition detected

5.3. Custom Colors

All indicator colors can be customized using the command protocol (see [Command Protocol](#)).

Table 2. Default LED Color States

Channel Type	State	Default Color	Indication
Landing Gear	DOWN	Green	Gear down and locked
Landing Gear	UP	Blue	Gear up and locked
Landing Gear	MOVING	Blinking Green/Blue	Gear in transition
Landing Gear	ERROR	Blinking Red	Fault detected
Rudder	DOWN	Yellow	Central position
Rudder	UP	Purple	Deflected position
Rudder	MOVING	Blinking Yellow/Purple	Position in transition
Rudder	ERROR	Blinking Red	Fault detected

Chapter 6. Test Switch Functions

The system includes a test switch that provides access to diagnostic and configuration functions:

6.1. Momentary Press

A brief press of the test switch during normal operation will initiate a self-test sequence:

- LED indicators will cycle through red, green, and blue
- System will test all relays
- Results will be displayed via LED indicators

6.2. Extended Press

Pressing and holding the test switch for more than 5 seconds enters LED brightness adjustment mode:

1. LEDs will display white at different brightness levels
2. Release the test switch when desired brightness is achieved
3. Setting is automatically saved to memory



The test switch is also used during firmware updates to enter the bootloader mode. See [Firmware Update](#) for details.

Chapter 7. Configuration Options

The system supports the following configuration options:

7.1. User Settings

- LED brightness
- LED colors for different states
- Wait time before operation

7.2. Channel Settings

For each channel:

- Channel type (landing gear or rudder)
- I2C addresses for monitoring and control devices
- Voltage limits (minimum and maximum)
- Calibration values for current sensing

Chapter 8. Command Protocol

The system supports a command-based protocol for configuration and diagnostics via the UART interface.

8.1. Supported Commands

- **v**: Get application version
- **r**: Reset device
- **s**: Scan I2C devices (parameter: I2C address)
- **u**: Get user settings
- **U**: Update user settings
- **c**: Get channel settings (parameter: channel number)
- **C**: Update channel settings
- **m**: Get monitoring data (parameter: channel number)
- **t**: Test specific channel

8.2. Command Format

Commands are sent as ASCII strings with the following format:

```
<command_char><optional_parameters>
```

Example:

```
v          # Get version
c2         # Get settings for channel 2
m0         # Get monitoring data for channel 0
```

Responses are formatted as JSON objects with command-specific data fields.

Example Command Response

```
{
  "command": "v",
  "status": "success",
  "data": {
    "version": "1.0.5",
    "build_date": "2025-03-15"
  }
}
```

Chapter 9. Troubleshooting

9.1. Error Indicators

Error conditions are indicated by red blinking LEDs. The specific pattern may indicate the error type:

- **Rapid blinking:** Communication error with I2C device
- **Slow blinking:** Power parameter out of range
- **Double blink:** Motor control failure

9.2. Common Issues

Problem	Possible Cause	Solution
System doesn't power up	Incorrect power connection	Check power polarity and voltage
Channel doesn't respond	Channel disabled in settings	Enable channel in configuration
Error indication on channel	Excessive current draw	Check for mechanical obstruction
Error indication on channel	Device communication error	Check I2C connections
Erratic operation	Electrical noise	Improve wiring shielding

9.3. Firmware Update

If firmware update is required:

1. Connect to UART interface
2. Put system in bootloader mode by power cycling while holding test switch
3. Use the firmware update tool to send new firmware
4. Restart the system after update completes



Do not interrupt the firmware update process once it has started. Doing so may render the device inoperable and require factory service.

Chapter 10. Maintenance

10.1. Routine Inspection

Perform the following checks regularly:

- Inspect all electrical connections for security
- Check for signs of overheating or damage
- Test system operation using the test switch
- Verify correct LED indication for all states



For experimental aircraft, maintenance of the control system should be included in your aircraft’s maintenance schedule and logbook. Regular testing is essential for ensuring continued safe operation.

10.2. Calibration

The system should be recalibrated annually or after any significant system modification:

1. Access the calibration function via the command protocol
2. Follow the prompted sequence to measure reference values
3. Save the new calibration data

Table 3. Recommended Maintenance Schedule

Interval	Maintenance Item	Description
Weekly	Visual inspection	Check for loose connections, damage, or overheating
Monthly	Functional test	Use test switch to verify proper operation
Annually	Calibration	Perform full system calibration
As needed	Firmware update	Update to latest firmware version

Chapter 11. Technical Specifications

11.1. Electrical

- Input voltage: 24V DC nominal (20-30V DC range)
- Power consumption: 2A maximum (all channels active)
- Channel output: 2A maximum per channel
- Communication: UART, 115200 baud

11.2. Mechanical

- Dimensions: 120mm x 80mm x 30mm
- Weight: 200g
- Operating temperature: -20°C to +70°C
- Storage temperature: -40°C to +85°C
- Humidity: 0-95% non-condensing

11.3. Compliance Information

- Designed for experimental and special category aircraft
- Not certified for installation in type-certificated aircraft
- Builder/operator responsible for compliance with applicable regulations



While this product is designed to high quality standards, it does not have formal certification for installation in type-certificated aircraft. Installation in certified aircraft would require appropriate field approval or STC process.

Chapter 12. Warranty and Support

The Digital Float Control System is warranted against defects in materials and workmanship for a period of two years from the date of purchase.

For technical support:

- Email: support@skymatikaero.com
- Phone: +1-555-123-4567
- Web: www.skymatikaero.com/support

Please have your serial number ready when contacting support.

Appendix A: Command Reference

This appendix provides a complete reference of all available commands for the Digital Float Control System.

Table 4. Command Summary

Command	Parameters	Description
v	None	Get application version information
r	None	Reset the device
s	I2C address	Scan for I2C device at specified address
u	None	Get current user settings
U	JSON settings object	Update user settings
c	Channel number (0-5)	Get settings for specified channel
C	JSON channel settings object	Update channel settings
m	Channel number (0-5)	Get monitoring data for specified channel
t	Channel number (0-5)	Test specified channel

Appendix B: Glossary

LED

Light Emitting Diode. Used for visual status indication.

UART

Universal Asynchronous Receiver/Transmitter. Serial communication interface.

I2C

Inter-Integrated Circuit. A serial communication bus used for device communication.

Bootloader

Software that loads when a device powers on, allowing firmware updates.

Experimental Aircraft

Aircraft built by amateurs, from kits, or for research and development purposes, not certified under standard type certificates.

Special Category Aircraft

Aircraft with special airworthiness certificates such as limited, exhibition, or amateur-built.

Colophon

This document was created using AsciiDoc and formatted for print and digital distribution.

Digital Float Control System User Manual for Experimental Aircraft
Version 1.0 (April 15, 2025)

Copyright © 2025 Skymatik Aero. All Rights Reserved.

Printed in the Republic of Poland

```
<style>
@page {
  @bottom-center {
    content: "Page " counter(page) " of " counter(pages);
    font-size: 9pt;
  }
  @bottom-left {
    content: "© 2025 Skymatik Aero";
    font-size: 9pt;
  }
  @bottom-right {
    content: "Digital Float Control Manual v{revnumber} - For Experimental Aircraft";
    font-size: 9pt;
  }
}
</style>
```